

How Can I Swing Like Pro?: Golf Swing Analysis Tool for Self Training

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ABSTRACT

In this work, we present an analysis tool to help golf beginners compare their swing motion to the swing motion of experts. The proposed application synchronizes videos with different swing phase timings using the latent features extracted by a neural network-based encoder and detects key frames where discrepant motions occur. We visualize synchronized image frames and 3D poses that help users recognize the differences and key factors that can be important for their swing skill improvement.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

computer vision, machine learning, motor skill training

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1 INTRODUCTION

In golf, correct swing forms lead to excellent outcomes. For golf beginners, a common way to understand and learn a correct swing motion naturally is to observe professional players' swings and imitate their forms. However, it is hard for beginners to specify key frames that they should focus on and which part of the body they should correct due to inconsistent timing and a lack of knowledge.

In this work, we propose a golf swing analysis tool using deep neural networks to help the user recognize the difference between the user's swing video and an expert's video obtained from the internet or broadcasted media. The proposed tool visualizes the image frames and human motions where the discrepancy between the expert and the user's swing is large, and helps the user easily grasp the motion that needs to be corrected.

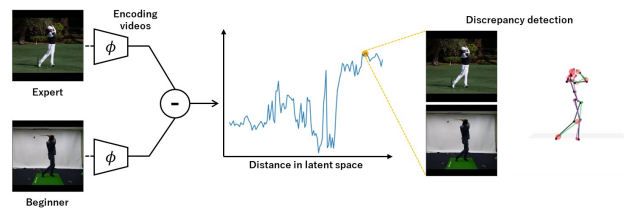


Figure 1: Overview of the proposed approach and visualized analysis results. Through the trained encoder, two input videos are synchronized and compared using the distance in the latent space. The large distance in the latent space is detected as a key frame and detailed difference between the two motion is shown.

2 OUR APPROACH

We create a prototype application that visualizes the distance of the latent space of two input videos, detects discrepant frames using an adaptive threshold, and compares the detected frames with 3D human poses (Figure 1). Since the phasing and timing of a golf swing can vary from person to person, we synchronize the input video clips using the Temporal Cycle-Consistency (TCC) network [Dwibedi et al. 2019]. The TCC network is a self-supervised learning algorithm; therefore, it can be trained with unlabeled data such as videos on the internet and broadcasted media and can generally fit many other types of motion analysis.

The inputs for the network are two video clips and the outputs of the network are two compressed latent vectors. The distance between the two latent vectors indicates the similarity of the two videos. This characteristic of the latent space can be used for synchronizing videos within the same category of motion. In this work, we focus on the distance between two videos in the latent space for detecting and retrieving fine-grained discrepancies to compare between two different swing forms, especially between beginners and experts.

For the application implementation, we first extract the latent vectors frame by frame using the TCC network and then synchronize the input videos by calculating the Euclidean distance between the latent vectors. At this point, similar motions should appear to be close in the latent space. However, repeated motions appear frequently during a golf swing, e.g., a particular posture can be observed both during the backswing and downswing phase. Therefore, we apply a dynamic time warping algorithm to penalize the distance in the latent space to gain better alignment.

To visualize the 3D poses of players, we use HRNet [Sun et al. 2019] and a simple linear network [Martinez et al. 2017], and then

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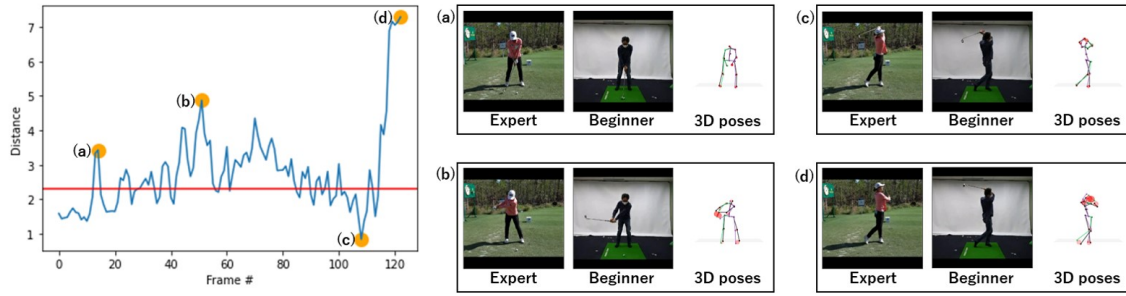


Figure 2: Comparison among the distance in the latent space, the two input images and the estimated 3D skeletons. The red line in the graph indicates the threshold for discrepancy detection. The colored skeleton and black skeleton indicate the user’s pose and expert’s pose, respectively. The intensity and radius of red circles indicate the degree of joint position difference of the two skeletons.

we apply Procrustes analysis to align two joint sets from different camera positions. This feature allows the users to see the motion difference between themselves and the expert and intuitively recognize which part they should correct to mimic the expert’s motion.

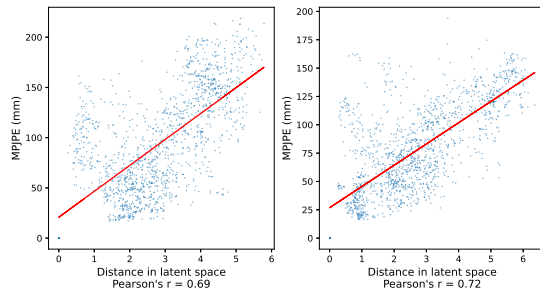


Figure 3: Correlation between the distance in the latent space and the MPJPE. Left: with club; Right: without club.

3 EVALUATION

As our video dataset, we use GolfDB [McNally et al. 2019] which consists of 1400 high-quality video clips of experienced golfers’ swings from online video platforms. To explore the effectiveness of the application, we investigate whether the distance in the latent space can represent the discrepancy between two input video frames. We calculate the mean per joint position error (MPJPE) of the two estimated 3D skeletons and compare it with the corresponding distance in the latent space. In the experiment, an over 0.69 Pearson’s correlation coefficient shows the correlation between the distance in the latent space and the MPJPE (Figure 3). This means that when the distance in the latent space is small, the difference between the two 3D poses is small (Figure 2.c) and when the distance is large, the 3D poses show more differences (Figure 2.b, d).

As depicted in Figure 2.a, we also observe that even in certain circumstances where the joint difference is small, the distance in the latent space remains large. This may be explained by the fact

that the network focuses on not only human motion, but also other motion features. In this case, the golf club’s movement, which is also considered critical during the swing, might be the main factor causing the large distance in the latent space. As shown in Figure 3, a greater correlation is found if the club was removed from the scene. This indicates that motion features of the club can be learned by the network, and it will be important to analyze the club motion when showing the difference between the two input motions.

Based on the results, we observe that the proposed tool can retrieve motion features that help users target the key moves that they have to correct. However, to achieve a more precise comparison between the latent distance and the motion difference, other potential motion features should also be considered in such circumstances.

4 CONCLUSION

In this work, we create an application for analyzing and visualizing the discrepancy of two input golf swing motions. Users can easily grasp the difference between their swings and various experts’ swings during self-training. Our early experiment confirms that the proposed tool can detect discrepancies between two swing videos. Furthermore, we provide an interpretation of the network that can help explain the correlation among the learned latent space, the human motion, and the other motion features. In the future, we plan to investigate more about the club’s motion and implement a system that proposes optimal golf swing corrections while considering the essential factors that directly affect the user’s golf swing performance.

REFERENCES

- D. Dwibedi, Y. Aytar, J. Tompson, P. Sermanet, and A. Zisserman. 2019. Temporal Cycle-Consistency Learning. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1801–1810. <https://doi.org/10.1109/CVPR.2019.00190>
- J. Martinez, R. Hossain, J. Romero, and J. J. Little. 2017. A Simple Yet Effective Baseline for 3d Human Pose Estimation. In *2017 IEEE International Conference on Computer Vision (ICCV)*. 2659–2668. <https://doi.org/10.1109/ICCV.2017.288>
- W. McNally, K. Vats, T. Pinto, C. Dulhanty, J. McPhee, and A. Wong. 2019. GolfDB: A Video Database for Golf Swing Sequencing. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 2553–2562. <https://doi.org/10.1109/CVPRW.2019.00311>
- K. Sun, B. Xiao, D. Liu, and J. Wang. 2019. Deep High-Resolution Representation Learning for Human Pose Estimation. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5686–5696. <https://doi.org/10.1109/CVPR.2019.00584>